

# Wirsching’s 2003 Conjectures on Positive Predecessor Density: A Proof, a Microcanonical Decomposition, and a Certified Numerical Test

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## Abstract

Wirsching (2003) reduces uniform positive predecessor density for the  $3n + 1$  map to a chain of five conditions, organized into three conjectures. Conjecture 1 concerns Wirsching’s path-counting generators. Conjecture 2 concerns a one-dimensional averaging operator  $W_3$ , whose unique fixed point of unit mass  $\varphi$  is a base-3 analogue of the Fabius function. Conjecture 3 concerns the asymptotics of  $\varphi$  itself. We prove Conjecture 1 by a generating-function cancellation. Conjecture 2’s hypothesis carries no information about the residue coordinate the rest of the chain needs: the primary source supplies two bridges toward it, neither uniform in growing resolution. We give a microcanonical decomposition of the Syracuse law in terms of Wirsching’s generators, prove exact equivalence-of-ensembles theorems at fixed and sublinear precision, and prove that the same decomposition’s latent cost vectors fail to converge at linear precision, a narrower fact than nonequivalence after projection, which remains open. The decomposition’s own open hypothesis, a uniform lower bound on the generators relative to their mean, remains open too. At the support boundary, an exhaustive fixed-integer test shows no deterioration through  $\ell = 17$ , while the population’s low quantiles decay at every offset tested past that boundary, a decay not confined to extreme-value statistics of the growing index set. A monotonicity certificate available for a related count in a companion paper does not transfer here. None of this closes Conjecture 2. An exact evaluation of  $\varphi$  through depth  $\ell = 500$ , zero truncation error at every point tested, supports Conjecture 3 for every  $u$  tested, with a central-sequence extrapolation coefficient stable to within  $4 \times 10^{-4}$  under the sub-range and model checks tested. This measurement cannot by itself distinguish a constant from a periodic factor multiplying the predicted asymptotic; a same-phase drift check disfavors a dominant periodic factor without ruling one out.

## 1 Introduction

Wirsching [1] studies uniform positive predecessor density for the Collatz map through a functional equation for a specific probability density  $\varphi$ , a base-3 analogue of the Fabius function, arising as the limiting density of a normalized path count in the Collatz reverse tree. The paper reduces the target density statement to a chain of implications through three increasingly concrete conjectures, of which only the first is settled, by §3 below; the third has numerical support below, and the second, on which the whole chain pivots, has remained open.

§2 fixes Wirsching’s objects and states the chain. §3 proves the first conjecture. §4 develops the microcanonical decomposition of the Syracuse law that connects Wirsching’s path counts to

the weighted covering formulation of a companion paper on the endogeny barrier [7]. It adds the equivalence- and nonequivalence-of-ensembles theorems, which bound how far local-limit methods reach toward the second conjecture, and an empirical separation of the two confounds behind the decomposition's own open uniform lower bound. §5 reports the certified numerical test of the third conjecture.

## 2 Wirsching's chain, from the primary source

Wirsching's central combinatorial objects are the “Elka” functions  $e_\ell(k, a) := |E_{\ell,k}(a)|$ , on domain  $\mathbb{N}_0 \times \mathbb{Z}_3^\times$  [1, §1–2], where  $E_{\ell,k}(a)$  is the set of paths  $b \xrightarrow{T} \cdots \xrightarrow{T} a$  in the Collatz graph with  $T = T_0$  exactly  $k$  times and  $T = T_1$  exactly  $\ell$  times ( $T_0, T_1$  the two branches of the accelerated  $3n+1$  map). Their normalized generators  $\tilde{g}_\ell$  (defined below) live on the continuous state space  $\mathbb{X} = \mathbb{I}_3 \times \mathbb{Z}_3^\times$ : the  $\mathbb{Z}_3^\times$  factor carries the 3-adic variable habitat of the Syracuse measure, while the  $\mathbb{I}_3 \cong \{0, 1, 2\}^\mathbb{N}$  factor carries the normalized step count  $k/3^\ell$ , converging to a density  $\varphi$ .

The compactly supported *generators*  $g_\ell(k, a)$  are defined by the recursion [1, §2, eq. (2.1)]

$$g_{\ell+1}(k, a) = \sum_{0 \leq j < 2 \cdot 3^\ell} g_\ell(k - j, \frac{2^{j+1}a-1}{3}), \quad g_0(k, a) = \mathbf{1}[k = 0] \mathbf{1}[a \in \mathbb{Z}_3], \quad (1)$$

linked to the Elka functions by  $e_\ell = p_\ell * g_\ell$ , where  $p_\ell(m)$  counts the ways to pay  $m$  with coins  $c_0 = 1$ ,  $c_j = 2 \cdot 3^{j-1}$  ( $j \geq 1$ ) [1, §2]. The class  $A_\delta$  consists of the integer sequences  $(k_\ell)$  with  $|\ell - k_\ell| \leq \delta \sqrt{\ell}$  for every  $\ell$  [1, eq. (1.5)];  $\tilde{A}_\delta$  is the analogous class of sequences  $(x_\ell) \in \mathbb{I}_3^\mathbb{N}$  [1, eq. (3.2)], used in condition (★4) below and in the numerical test of §5.  $\varphi$  itself satisfies the untruncated equation  $\lambda \varphi'(t) = a(\varphi(at) - \varphi(at - a + 1))$  for  $a = 3$ ,  $\lambda = 2/3$  [1, eq. (7.9)].  $\varphi_0$  is Berg and Krüppel's explicit asymptotic [2, Proposition 9.1, eq. (9.6)] for a particular solution of the truncated equation  $\lambda g'(t) = ag(at)$ , a four-constant closed form of the shape  $t^{(\cdot)}(-\ln t)^{(\cdot)} \exp(-(\cdot) \ln^2(t/(-\ln t)))$  whose constants  $\beta, \gamma, \delta, \varepsilon$ , evaluated at  $a = 3$ ,  $\lambda = 2/3$  (their own notation, kept separate here to avoid clashing with  $\delta$  elsewhere in this paper, and depending on one auxiliary constant not itself appearing in the closed form), are reproduced verbatim in the accompanying repository (§5 below); Berg and Krüppel expect only that  $\varphi/\varphi_0$  stays bounded and bounded away from zero, weaker than convergence [1, eq. (7.10)–(7.11)];  $\varphi_0$  is used in Conjecture 3 below in place of  $\varphi$  itself, which has no known closed form.

**Proposition 2.1** (Base-3 Fabius function).  *$\varphi$  is the density of  $X = \sum_{j \geq 1} 2U_j \cdot 3^{-j}$ ,  $U_j$  i.i.d. uniform on  $[0, 1]$  (an identification not in the source), the base-3 analogue of the Fabius density (the base-2 case satisfies  $f'(x) = 4f(2x)$  on  $[0, 1/2]$ ; here  $\varphi'(x) = \frac{9}{2}\varphi(3x)$  on  $[0, 2/3]$  by [1, eq. (7.7)], the same shape with base 3 in place of base 2: coefficient  $4 = 2^2/(2-1)$  becomes  $9/2 = 3^2/(3-1)$ , support cutoff  $1/2 = (2-1)/2$  becomes  $2/3 = (3-1)/3$ ). Among functions with  $\text{supp } \varphi \subset [0, 1]$  and  $\int_0^1 \varphi = 1$ , it is the unique fixed point of the averaging operator  $W_3 f(x) := \frac{3}{2} \int_{3x-2}^{3x} f(t) dt$ ,  $C^\infty$ , piecewise polynomial away from the standard Cantor set [1, Corollary 7]. For  $f_0$  with  $\int_0^1 f_0 = 1$ , the iterates  $f_{n+1} := W_3 f_n$  converge geometrically,  $\|f_n - \varphi\|_1 \leq 2^{-n+1} \|f_1 - f_0\|_1$  [1, Corollary 8].*

Wirsching's paper reduces uniform positive predecessor density (that the sets of predecessors of  $a$  have lower asymptotic density bounded below by  $c/a$ , for a single constant  $c > 0$  uniform over

admissible targets  $a$ ) to a chain of five conditions [1, Definition 1, §§1–7]:

|                           |                                                 |
|---------------------------|-------------------------------------------------|
| Theorem 1 (proved in [1]) | (★1) $\Rightarrow$ Positive Density             |
| Conjecture 1              | (★2) $\Rightarrow$ (★1)                         |
| Theorem 2 (proved in [1]) | (★3) $\Rightarrow$ (★2)                         |
| Conjecture 2              | (★4) $\Rightarrow$ (★3)                         |
| Conjecture 3              | (★5); (★5) $\Rightarrow$ (★4) proved in [1, §7] |

where (★1) is a  $\liminf$  lower bound on Wirsching's Elka functions relative to their Haar average, uniform over sequences  $(k_\ell)$  with  $|\ell - k_\ell| \leq \delta\sqrt{\ell}$ ; (★2) is the same statement for his generators  $g_\ell(k, a)$ ; (★3) upgrades the  $\liminf$  over  $\ell$  to a bound holding for every  $\ell \geq \ell_0$ ; and (★4) is

$$\liminf_{\ell} \frac{(W_3^\ell \chi_1)(x_\ell)}{(W_3^\ell \chi_0)(x_\ell)} \geq \mu, \quad \text{uniformly for } (x_\ell) \in \tilde{A}_\delta,$$

with  $\chi_0 = \mathbf{1}_{[0,2/3]}$ ,  $\chi_1 = \mathbf{1}_{[1/3,1]}$ , and  $W_3$  the averaging operator of Proposition 2.1.

First, (★4) is a statement purely about the one-dimensional operator  $W_3$ : it never mentions Wirsching's generators, and carries no information about the residue coordinate that (★3) needs. The source offers two bridges: the averaged-generator identity  $\lim_{\ell} \int_{\mathbb{Z}_3^\times} \tilde{g}_\ell(x, a) da = \varphi(x)$  for fixed  $x \in \mathbb{I}_3$  [1, §6], and Wirsching's Theorem 3 [1, §5], strong convergence  $S_\ell \rightarrow S_\infty$  for every  $f \in C(\mathbb{X})$ , of the transition operators  $S_\ell : C(\mathbb{X}) \rightarrow C(\mathbb{X})$  defined by  $\tilde{g}_\ell = S_\ell(\tilde{g}_{\ell-1})$  [1, §4], uniform on bounded equicontinuous families. Neither bridge is uniform in growing resolution, so neither closes the gap between (★4) and (★3).

Second, (★3) and (★2) are the same inequality. Writing  $\tilde{g}_\ell(x, a) = \gamma_\ell g_\ell(\lfloor 3^\ell x \rfloor, a)$  for a constant  $\gamma_\ell$  not depending on  $a$ , the constant cancels from both sides of the ratio each condition bounds; what Wirsching's Theorem 2 adds is a change of quantifier, from a  $\liminf$  in  $\ell$  to a bound holding at every  $\ell \geq \ell_0$ .

Third, Theorem 1's conclusion, uniform positive predecessor density, is about  $a \in \mathbb{N}$ , non-cyclic,  $a \not\equiv 0 \pmod{3}$ ; its hypothesis (★1), like (★2) and (★3), is stated over  $\mathbb{Z}_3^\times$ .

### 3 Wirsching's Conjecture 1

**Theorem 3.1** (Wirsching's Conjecture 1). *In the notation of §2, condition (★2), with window radius  $\delta_1$  (Wirsching's own naming, [1, §2]), implies condition (★1) with some possibly smaller window radius  $\delta \leq \delta_1$ .*

*Proof.* Write  $\bar{g}_\ell(k) := \frac{1}{2 \cdot 3^{\ell-1}} \sum_b g_\ell(k, b)$  and  $\bar{e}_\ell(k) := \frac{1}{2 \cdot 3^{\ell-1}} \sum_b e_\ell(k, b)$  for the residue-averages of the generators and the Elka functions. Put  $c_0 = 1$  and  $c_j = 2 \cdot 3^{j-1}$  for  $j \geq 1$ . If  $q_\ell(k)$  counts the bounded urn distributions defining  $\bar{g}_\ell(k)$  (urn  $j$  holds between 0 and  $c_j - 1$  balls, coefficient sum  $k$ ), then

$$\sum_{k \geq 0} q_\ell(k) z^k = \prod_{j=0}^{\ell} \frac{1 - z^{c_j}}{1 - z}, \quad \sum_{m \geq 0} p_\ell(m) z^m = \prod_{j=0}^{\ell} (1 - z^{c_j})^{-1}.$$

Since  $q_\ell(k) = 2 \cdot 3^{\ell-1} \bar{g}_\ell(k)$  and  $\bar{e}_\ell = p_\ell * \bar{g}_\ell$ , cancellation gives

$$\bar{e}_\ell(k) = \frac{1}{2 \cdot 3^{\ell-1}} \binom{k + \ell}{\ell}.$$

The unrestricted partition count for the coins  $1, 2, 6, 18, \dots$  is at most  $\exp(C \log^2(m+2))$ . Also

$$\frac{\binom{k-m+\ell}{\ell}}{\binom{k+\ell}{\ell}} \leq \left( \frac{k}{k+\ell} \right)^m.$$

Since  $p_\ell(0) = 1$ ,  $\bar{g}_\ell(j) \leq \bar{e}_\ell(j)$  for every  $j$ , so, relative to  $\bar{e}_\ell(k)$ , the part of the convolution  $\bar{e}_\ell(k) = \sum_m p_\ell(m) \bar{g}_\ell(k-m)$  with  $m \geq \tau\sqrt{\ell}$  is, uniformly for  $|k-\ell| \leq \delta\sqrt{\ell}$ , at most  $\exp(-c\sqrt{\ell} + O(\log^2 \ell))$ , for a tail-window radius  $\tau > 0$ . For  $k$  in that window and  $m < \tau\sqrt{\ell}$ ,  $|k-m-\ell| \leq (\delta+\tau)\sqrt{\ell}$ ; choosing  $\delta+\tau < \delta_1$  puts  $j = k-m$  inside the window of  $(\star 2)$ , and positivity transfers its lower bound from  $g_\ell/\bar{g}_\ell$  to  $e_\ell/\bar{e}_\ell$ .  $\square$

*Remark 3.2* (Relation to the source). Wirsching records the partition-counting identity  $\prod_{j=0}^\ell (1 - z^{c_j})^{-1} = \sum_m p_\ell(m) z^m$  [1, §2, eq. (2.2)] and the closed form for  $\bar{e}_\ell(k)$  used above [1, §1, eq. (1.4)], and describes, without a generating function, the urn count  $q_\ell(k) = 2 \cdot 3^{\ell-1} \bar{g}_\ell(k)$ , crediting the combinatorial identification to [3] [1, §2]; turning that description into the generating function  $\sum_k q_\ell(k) z^k = \prod_{j=0}^\ell (1 - z^{c_j}) / (1 - z)$  used above is this proof's own step. Wirsching states the implication itself as Conjecture 1 [1, §§1–2]. The cancellation  $P_\ell(z) Q_\ell(z) = (1-z)^{-\ell-1}$  behind  $\bar{e}_\ell(k)$  is checked coefficient by coefficient in the accompanying repository; the convolution-tail estimate and the window choice  $\delta+\tau < \delta_1$  that close the implication are analytic and checked by hand against the primary source.

The second conjecture remains open. The most concrete sufficient condition above it is Conjecture 3, an asymptotic assertion on  $\varphi(z_\ell)/\varphi_0(z_\ell)$  along the central-limit window  $|k-\ell| \leq \delta\sqrt{\ell}$ , which we test numerically in §5.

## 4 A microcanonical decomposition, and how far local-limit methods reach

**Theorem 4.1** (Microcanonical decomposition). *Let  $g_\ell(k, a)$  be Wirsching's generator, let  $c_i = 2 \cdot 3^i$  for  $i \geq 0$  (the coin  $c_j$  of §3 reindexed by  $i = j-1$ , dropping the extra coin  $c_0 = 1$  that appears there only), and let  $\mu_\ell$  be the Syracuse law modulo  $3^\ell$ : the law of the residue built by the same i.i.d. geometric branching used in the proof below, matching Tao's Syracuse random variable [6] (unrelated to the constant  $\mu$  of  $(\star 1)$ ,  $(\star 3)$ ,  $(\star 4)$  in §2). For every unit residue  $a$ ,*

$$\mu_\ell(a) = \left( \prod_{i=0}^{\ell-1} \frac{1}{2(1-2^{-c_i})} \right) \sum_{k=0}^{3^\ell-1} 2^{-k} g_\ell(k, a). \quad (2)$$

Suppose there are  $\delta, \eta > 0$  such that, for all sufficiently large  $\ell$ , every unit  $a$ , and every integer  $|k-\ell| \leq \delta\sqrt{\ell}$ ,

$$g_\ell(k, a) \geq \eta \bar{g}_\ell(k), \quad \bar{g}_\ell(k) = \frac{1}{2 \cdot 3^{\ell-1}} \sum_{b \in (\mathbb{Z}/3^\ell \mathbb{Z})^\times} g_\ell(k, b).$$

Then  $\mu_\ell(a) \geq C_{\delta, \eta} 3^{-\ell}$  uniformly in  $a$  and  $\ell$ . In particular, condition  $(\star 3)$  implies a uniform lower bound on every cylinder of the Syracuse law, the estimate the weighted covering formulation of the companion paper on the endogeny barrier needs, that the total geometric multiplicity of a residue's representations stays within a subexponential factor of its mean, uniformly over residues [7]. The hypothesis  $g_\ell(k, a) \geq \eta \bar{g}_\ell(k)$  is open: Empirical Result 4.9 below finds the minimum multiplicity is zero at  $k = \ell$  at every level computed, so no  $\eta > 0$  works there yet. Since  $k_\ell = \ell$  lies in Wirsching's

own window too, a zero at  $k = \ell$  recurring at arbitrarily large  $\ell$  would falsify this hypothesis for every  $\eta > 0$ , and falsify both  $(\star 2)$  and  $(\star 3)$  for every choice of Wirsching's own constants there: a  $\liminf$ -type hypothesis tolerates only finitely many exceptions, so persistence through every level computed so far does not by itself decide the question. Whether the zero set at  $k = \ell$  is eventually empty or persists indefinitely is exactly what a finite computation cannot settle.

*Proof.* At recursion level  $i$ , multiplication by 2 has order  $c_i = 2 \cdot 3^i$  modulo  $3^{i+1}$ . If  $\Gamma_i$  has the geometric law  $\Pr(\Gamma_i = m) = 2^{-m}$  for  $m \geq 1$ , then  $J_i = \Gamma_i - 1 \bmod c_i$  satisfies

$$\Pr(J_i = j) = \frac{2^{-(j+1)}}{1 - 2^{-c_i}}, \quad 0 \leq j < c_i.$$

Folding each exponent independently by the applicable order leaves the Syracuse residue unchanged: from the recursion  $a = (3a' + 1) \cdot 2^{-(j+1)}$ , the residue  $a \bmod 3^{i+1}$  is a function of  $a' \bmod 3^i$  and of  $j \bmod c_i$  alone, since  $c_i = 2 \cdot 3^i$  is exactly the order of 2 modulo  $3^{i+1}$ . Wirsching's recursion for  $g_\ell$  counts the folded vectors with  $\sum_i J_i = k$ . Multiplying their probabilities gives (2).

Put  $K_\ell = \sum_{i < \ell} J_i$ . The summands are independent and are uniformly dominated by an untruncated geometric variable. Direct summation gives  $\mathbb{E}K_\ell = \ell + O(1)$  and  $\text{Var } K_\ell = 2\ell + O(1)$ . The Lindeberg central limit theorem therefore gives

$$\Pr(|K_\ell - \ell| \leq \delta\sqrt{\ell}) \longrightarrow 2\Phi(\delta/\sqrt{2}) - 1 > 0.$$

Sum (2) over the central values of  $k$  and use the assumed lower bound. Since there are  $2 \cdot 3^{\ell-1}$  unit residues, this yields

$$\mu_\ell(a) \geq \frac{\eta}{2 \cdot 3^{\ell-1}} \Pr(|K_\ell - \ell| \leq \delta\sqrt{\ell}) \geq C_{\delta,\eta} 3^{-\ell}.$$

□

**Proposition 4.2** (Conditional local-limit bridge). *Set*

$$G_{\ell,a}(z) = \sum_k g_\ell(k, a) z^k, \quad \overline{G}_\ell(z) = \frac{1}{2 \cdot 3^{\ell-1}} \sum_a G_{\ell,a}(z).$$

Define the conditional cost laws

$$Q_{\ell,a}(k) = \frac{2^{-k} g_\ell(k, a)}{G_{\ell,a}(1/2)}, \quad \overline{Q}_\ell(k) = \frac{2^{-k} \overline{g}_\ell(k)}{\overline{G}_\ell(1/2)}.$$

Then the exact identity

$$\frac{g_\ell(k, a)}{\overline{g}_\ell(k)} = \frac{G_{\ell,a}(1/2)}{\overline{G}_\ell(1/2)} \frac{Q_{\ell,a}(k)}{\overline{Q}_\ell(k)} \tag{3}$$

holds. Fix  $\delta > 0$ . If, uniformly in unit  $a$  and  $|k - \ell| \leq \delta\sqrt{\ell}$ ,

$$\frac{G_{\ell,a}(1/2)}{\overline{G}_\ell(1/2)} \geq \eta > 0, \quad Q_{\ell,a}(k) \geq c_\delta \ell^{-1/2},$$

then condition  $(\star 3)$  holds. Via (2),  $G_{\ell,a}(1/2)/\overline{G}_\ell(1/2) = 2 \cdot 3^{\ell-1} \mu_\ell(a)$ , so the first hypothesis is exactly Theorem 4.1's conclusion  $\mu_\ell(a) \geq C 3^{-\ell}$ . The proposition packages that conclusion with a second, local-limit hypothesis. The second hypothesis is open at  $k = \ell$  for the same reason as in Theorem 4.1: the minimum computed multiplicity is zero at every level tested. The first is open because the only available lower bound on the canonical mass ratio, the conjectural weighted-covering estimate discussed below, is only subexponential.

*Proof.* Identity (3) follows by cancellation. Under the folded geometric law in the proof of Theorem 4.1,  $\overline{Q}_\ell$  is the law of  $K_\ell$ . The summands have uniformly bounded exponential moments, mean sum  $\ell + O(1)$ , and variance  $2\ell + O(1)$ . Their lattice local central limit theorem gives, uniformly in the stated window,

$$\overline{Q}_\ell(k) \asymp_\delta \ell^{-1/2}.$$

The two assumed lower bounds and (3) now give a positive uniform lower bound for  $g_\ell(k, a)/\overline{g}_\ell(k)$ .  $\square$

The first factor in (3) is the canonical Syracuse mass relative to its mean. The conjectural weighted-covering estimate gives only a subexponential lower bound for this factor; the proposition assumes a constant lower bound. The second factor is a local limit theorem after conditioning on the residue. Its Fourier representation is

$$Q_{\ell,a}(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ikt} \frac{G_{\ell,a}(e^{it}/2)}{G_{\ell,a}(1/2)} dt.$$

*Remark 4.3* (Local limit theorems for operator cocycles). Spectral local limit theorems exist for random expanding dynamics [4] and for time-dependent expanding systems [5]. Their hypotheses include a twisted operator cocycle acting on a fixed Banach space, a uniform spectral or Ruelle–Perron–Frobenius gap near zero, and an aperiodicity estimate away from zero. These hypotheses would give the phase control in Proposition 4.2. They have not been verified for Wirsching’s  $S_\ell$ . Theorem 3 of [1] gives strong convergence for every  $f$ , uniform only on fixed equicontinuous families, while the densities used here acquire resolution  $3^{-\ell}$ . The available local limit theorems do not remove that growing-resolution obstruction.

**Theorem 4.4** (Equivalence of ensembles at fixed precision). *For a unit residue  $a$  put*

$$p_{\ell,k}(a) = \frac{g_\ell(k, a)}{\sum_b g_\ell(k, b)}.$$

*Fix  $r \geq 1$  and  $\delta > 0$ . Uniformly over integers  $|k - \ell| \leq \delta\sqrt{\ell}$ , the projection of  $p_{\ell,k}$  modulo  $3^r$  converges in total variation to the Syracuse law  $\mu_r$  as  $\ell \rightarrow \infty$ .*

*Proof.* Under the canonical weights in Theorem 4.1, the folded costs  $J_0, \dots, J_{\ell-1}$  are independent and  $p_{\ell,k}$  is the law of the resulting residue conditional on  $K_\ell = \sum_i J_i = k$ . Fix the terminal block of  $r$  costs. For each fixed value  $\mathbf{j}$  of that block, with coordinate sum  $s(\mathbf{j})$ , independence gives

$$\Pr(\mathbf{J} = \mathbf{j} \mid K_\ell = k) = \Pr(\mathbf{J} = \mathbf{j}) \frac{\Pr(K_{\ell-r} = k - s(\mathbf{j}))}{\Pr(K_\ell = k)}.$$

The lattice local central limit theorem used in Proposition 4.2 makes the ratio tend to one, uniformly in the stated window, for every fixed  $\mathbf{j}$ . The geometric domination of the coordinates makes the block tails uniformly summable. Splitting at block sum  $o(\sqrt{\ell})$  and applying the same local estimate there extends the convergence to total variation on the whole block.

The terminal  $r$  folded costs determine the final residue modulo  $3^r$ : pairing  $J_{\ell-1-i}$ , for  $i = 0, \dots, r-1$ , against the reduced cap  $2 \cdot 3^{r-1-i}$  is valid because the original cap  $c_{\ell-1-i} = 2 \cdot 3^{\ell-1-i}$  is a multiple of  $2 \cdot 3^{r-1-i}$  for every  $r \leq \ell$ , which is exactly the divisibility  $d \mid C$  used below. At a terminal stage whose original cap is  $C$ , reduction to the cap  $d = 2 \cdot 3^i$  required at precision  $r$  is exact because  $d \mid C$  and

$$\sum_{\substack{0 \leq j < C \\ j \equiv s \pmod{d}}} \frac{2^{-(j+1)}}{1 - 2^{-C}} = \frac{2^{-(s+1)}}{1 - 2^{-d}}, \quad 0 \leq s < d.$$

The reduced block therefore has exactly the canonical folded-geometric law defining  $\mu_r$ . Total variation cannot increase under the deterministic map from the block to its residue, which proves the claim.  $\square$

**Theorem 4.5** (Equivalence of ensembles at sublinear precision). *Let  $r = r_\ell$  be an integer sequence with  $1 \leq r \leq \ell$  and  $r = o(\ell)$ . Fix  $\delta > 0$ . Uniformly over integers  $|k - \ell| \leq \delta\sqrt{\ell}$ , the projection of  $p_{\ell,k}$  modulo  $3^r$  has total-variation distance  $o(1)$  from  $\mu_r$ .*

*Proof.* It remains to treat  $r \rightarrow \infty$ , since bounded subsequences follow from Theorem 4.4. Split the canonical cost as  $K_\ell = R_{\ell,r} + B_{\ell,r}$ , where  $B_{\ell,r}$  is the sum of the terminal  $r$  folded costs. The Radon derivative of the law of the terminal block conditional on  $K_\ell = k$ , relative to its unconditioned law, is

$$D_{\ell,r}(B_{\ell,r}) = \frac{\Pr(R_{\ell,r} = k - B_{\ell,r})}{\Pr(K_\ell = k)}. \quad (4)$$

Choose  $h_\ell \rightarrow \infty$  slowly enough that  $h_\ell\sqrt{r/\ell} \rightarrow 0$ . Uniform exponential moments give

$$\Pr(|B_{\ell,r} - \mathbb{E}B_{\ell,r}| > h_\ell\sqrt{r}) = o(1).$$

On the complementary event, removing the block changes the centered local-CLT coordinate by  $o(1)$  and changes the variance by a relative  $o(1)$ . The uniform lattice local central limit theorem for the folded geometric triangular array therefore gives, in (4),  $D_{\ell,r}(B_{\ell,r}) = 1 + o(1)$  uniformly there. In the central window,  $\Pr(K_\ell = k) \gg_\delta \ell^{-1/2}$ , while the standard concentration bound for lattice sums with uniformly bounded exponential moments gives  $\sup_m \Pr(R_{\ell,r} = m) \ll (\ell - r)^{-1/2}$ . Hence  $D_{\ell,r} \ll_\delta 1$  on its whole support. Truncation of the exceptional set now gives  $\mathbb{E}|D_{\ell,r} - 1| = o(1)$ . This is twice the total-variation distance between the conditioned and unconditioned terminal blocks. Their deterministic residue maps cannot increase that distance. The cap-reduction identity in the proof of Theorem 4.4 shows that the unconditioned residue has law exactly  $\mu_r$ .  $\square$

**Theorem 4.6** (Linear-block nonequivalence). *Let  $\mathbf{B}_{\ell,r} = (J_{\ell-r}, \dots, J_{\ell-1})$  be the terminal block of folded costs and let  $B_{\ell,r} = \sum_{i=\ell-r}^{\ell-1} J_i$  be its coordinate sum, the same block sum used in the proof of Theorem 4.5. Suppose that*

$$\frac{r}{\ell} \longrightarrow \rho \in (0, 1), \quad \frac{k_\ell - \ell}{\sqrt{\ell}} \longrightarrow u.$$

*Then*

$$\begin{aligned} & d_{\text{TV}}(\mathcal{L}(\mathbf{B}_{\ell,r} \mid K_\ell = k_\ell), \mathcal{L}(\mathbf{B}_{\ell,r})) \\ &= d_{\text{TV}}(\mathcal{L}(B_{\ell,r} \mid K_\ell = k_\ell), \mathcal{L}(B_{\ell,r})) \\ &\longrightarrow d_{\text{TV}}(\mathcal{N}(\rho u, 2\rho(1 - \rho)), \mathcal{N}(0, 2\rho)) > 0, \end{aligned}$$

*where the second parameter of  $\mathcal{N}$  is the variance. The equality above is exact at every finite level; only the convergence that follows it is asymptotic. For  $u = 0$ , the limit equals*

$$2 \left[ \Phi \left( \sqrt{\frac{-\log(1 - \rho)}{\rho}} \right) - \Phi \left( \sqrt{\frac{-(1 - \rho) \log(1 - \rho)}{\rho}} \right) \right].$$

*Proof.* Write  $R_{\ell,r} = K_\ell - B_{\ell,r}$ . Relative to the unconditioned block law, the conditioned law has density

$$\frac{\Pr(R_{\ell,r} = k_\ell - B_{\ell,r})}{\Pr(K_\ell = k_\ell)}.$$

This density is a function of  $B_{\ell,r}$  alone. Taking its absolute deviation from one proves the finite-level equality of the two total variation distances.

The folded costs have uniformly bounded exponential moments. Their means and variances differ from 1 and 2, respectively, by summable errors. The lattice local central limit theorem, applied separately to  $B_{\ell,r}$  and  $R_{\ell,r}$ , gives the unconditioned limit  $\mathcal{N}(0, 2\rho)$  for  $(B_{\ell,r} - r)/\sqrt{\ell}$ . Write  $\psi_{m,\sigma^2}$  for the  $\mathcal{N}(m, \sigma^2)$  density, distinct from  $\varphi$  and  $\varphi_0$  above. After conditioning, its limiting density is

$$x \mapsto \frac{\psi_{0,2\rho}(x)\psi_{0,2(1-\rho)}(u-x)}{\psi_{0,2}(u)} = \psi_{\rho u, 2\rho(1-\rho)}(x).$$

Uniform exponential tails promote the local convergence to convergence in total variation. The two limiting variances are different, so the distance is positive. When  $u = 0$ , the two centered Gaussian densities cross at  $|x| = \sqrt{-2(1-\rho)\log(1-\rho)}$ ; integration on the central interval gives the displayed formula.  $\square$

Theorem 4.6 shows that the condition  $r = o(\ell)$  in Theorem 4.5 is sharp for the latent cost vectors. It does not prove nonequivalence after projection modulo  $3^r$ : the residue map can discard the block sum, and total variation can decrease under that projection. The linear-precision residue and Fourier problems remain open.

**Theorem 4.7** (Full-precision likelihood domination). *Let  $p_{\ell,k}^{(r)}$  be the projection of  $p_{\ell,k}$  modulo  $3^r$ , where  $1 \leq r \leq \ell$ . For every set  $E \subseteq \mathbb{Z}/3^r\mathbb{Z}$ ,*

$$p_{\ell,k}^{(r)}(E) \leq \Pr(K_\ell = k)^{-1} \mu_r(E). \quad (5)$$

Consequently, writing  $D_\infty(P\|Q) := \log \sup_x P(x)/Q(x)$  for the Rényi  $\infty$ -divergence,

$$D_{\text{KL}}(p_{\ell,k}^{(r)}\|\mu_r) \leq D_\infty(p_{\ell,k}^{(r)}\|\mu_r) \leq -\log \Pr(K_\ell = k).$$

Uniformly for  $|k - \ell| \leq \delta\sqrt{\ell}$ , the last quantity is  $\frac{1}{2} \log \ell + O_\delta(1)$ .

*Proof.* On the canonical folded-cost space, the microcanonical law is the conditional law given  $K_\ell = k$ . Hence, for the inverse image  $\tilde{E}$  of  $E$  under the residue map,

$$\Pr(\tilde{E} \mid K_\ell = k) \leq \frac{\Pr(\tilde{E})}{\Pr(K_\ell = k)}.$$

The unconditioned residue modulo  $3^r$  has law  $\mu_r$ , by the same cap reduction used in Theorem 4.4. This proves (5). It is the asserted bound on the essential likelihood ratio, and relative entropy is bounded by its logarithm. The lattice local central limit theorem gives  $\Pr(K_\ell = k) \asymp_\delta \ell^{-1/2}$  in the central window.  $\square$

Theorem 4.7 applies at full precision, but its direction is insufficient for condition  $(\star 3)$ . It proves that the relative entropy per digit vanishes and rules out an exponential likelihood separation from the canonical ensemble. Condition  $(\star 3)$  asks for a lower bound on every residue, so the unresolved part is the lower tail of the likelihood ratio.

For  $r = o(\ell)$  and  $\xi_0 \in \mathbb{Z}/3^r\mathbb{Z}$ , Theorem 4.5 gives

$$\hat{p}_{\ell,k}(3^{\ell-r}\xi_0) = \hat{\mu}_r(\xi_0) + o(1), \quad (6)$$

uniformly in  $\xi_0$  and in the central cost window. By (6), the fixed-cost law does not approach the uniform law on units at fixed precision. Fixed-conductor modes may persist and differ from

those of the uniform law itself:  $u_1$ , uniform on the two units modulo 3, has  $\widehat{u}_1(1) = -1/2$ , while  $\mu_1$ , which assigns masses  $1/3$  and  $2/3$  to the same two units, has  $\widehat{\mu}_1(1) = -\frac{1}{2} - \frac{i}{2\sqrt{3}}$ , modulus  $1/\sqrt{3} \neq 1/2$ . Any Fourier estimate for condition  $(\star 3)$  must separate these exact coarse modes and control frequencies whose conductor exponent is comparable with  $\ell$ .

**Empirical Result 4.8** (Fixed-precision projection test). *We projected the exact fixed-cost tables onto  $3^r$ , for  $r = 1, 2, 3$ , and compared them in total variation with  $\mu_r$  obtained from the independent Syracuse recursion. At  $\ell = 12$  and  $k = \ell$ , the distances were 0.02050, 0.04024, and 0.06228. At  $k = \ell + \lfloor \sqrt{\ell} \rfloor$ , they were 0.04310, 0.07902, and 0.12677. At fixed  $r$  and  $k = \ell$ , the distance decreases monotonically over every level tested,  $\ell = 3, \dots, 13$ . This finite test checks the implementation and is not used in the proof. At full precision and  $\ell = 13$ , the distances were 0.25195 at  $k = \ell$  and 0.26078 at  $k = \ell + \lfloor \sqrt{\ell} \rfloor$ . The corresponding precision-one values were 0.01885 and 0.04003. These full-precision values do not determine an asymptotic limit.*

**Empirical Result 4.9** (Finite microcanonical coverage). *We computed the multiplicities  $g_\ell(k, a)$  exactly through  $\ell = 12$ , checking at every level that their residue sum equals the independent bounded-composition count. At  $\ell = 12$ , the fractions of unit residues reached at  $k = \ell - \lfloor \sqrt{\ell} \rfloor, \ell, \ell + \lfloor \sqrt{\ell} \rfloor$  are respectively 0.2735, 0.8074, and 0.9973. The minimum multiplicity is still zero in all three cases. A separate Boolean recursion extends the support calculation through  $\ell = 16$ . At that level the central and upper-window coverage fractions are 0.9194 and 0.99999972, while the first fixed cost covering every residue equals  $\ell + 5$  for each  $10 \leq \ell \leq 16$ . These values neither prove nor refute the uniform asymptotic lower bound.*

**Empirical Result 4.10** (Separating extreme-value statistics from tail decay). *The minimum in Theorem 4.1's hypothesis and in Empirical Result 4.9 ranges over  $2 \cdot 3^{\ell-1}$  units, a set that grows with  $\ell$ . Its decay could be extreme-value statistics of that growth, or real deterioration at some fixed integer  $a$ . Two measurements bear on the question.*

*Over the 1458 units below  $3^7$ , an exhaustive, non-growing set of integers,  $\min_a g_\ell(\ell+5, a)/\bar{g}_\ell(\ell+5)$  reads 0.1019, 0.0736, 0.1073, 0.0787, 0.1161, 0.1075, 0.0958, 0.0892 for  $\ell = 10, \dots, 17$ , from a third recursion, restricted to this fixed set of integers rather than every unit residue, which reaches one level past the Boolean support recursion above. Eight levels, consistent with no trend; the ordinary least-squares slope is an order of magnitude below the series' own spread. The minimum over all units at the same offset reads 0.0509, 0.0368, 0.0536, 0.0197, 0.0435, 0.0215, 0.0240, 0.0297; its index set grows by a factor 3 at every level. This comparison is confined to  $d = +5$ : the same exhaustive test at the offsets  $d = 6, \dots, 12$  below, where the population as a whole does decay, has not been run.*

*A fixed low quantile of the population is immune to the extreme-order-statistic effect of a growing index set by construction: the value below which a fixed fraction of the population falls does not shrink just because the population grows. It is not immune to a change in the population's composition, which a growing index set can also produce and which this measurement does not separate out. At  $k = \ell + 5$  the  $q = 10^{-3}$  quantile of  $g_\ell(k, a)/\bar{g}_\ell(k)$  over units  $a$  stays in  $[0.10, 0.12]$  for  $\ell = 10, \dots, 16$ . At every other offset tested in Wirsching's window,  $d = 6, \dots, 12$ , the same quantile decays at a rate of 0.93–0.96 per level. At  $d = +12$ , for instance, it falls from 0.4326 to 0.2877 over the same range (0.93 per level), while the median falls from 1.0108 to 0.9528, so bulk drift accounts for only part of that fall. The offset  $d = +5$  is the first with complete support at every level (Empirical Result 4.9); its low quantile is dominated by residues whose count has only just left zero, a regime distinct from  $d \geq 6$ , where support has been complete for several levels already.*

*From  $d = +7$  on, the population's low quantiles decay at a rate close to the minimum's own (0.93–0.94 per level for both,  $\ell = 10, \dots, 16$ ; at  $d = +12$ , for instance, the minimum falls from 0.3840*

at  $\ell = 10$  to 0.2378 at  $\ell = 17$ , a rate of 0.93 per level, matching the quantile's own 0.93 above), even though the quantile itself sits well above the extreme order statistic, the single minimum. At  $d = +6$ , the offset immediately past the support boundary, the two diverge: the minimum does not decay at all over  $\ell = 10, \dots, 16$ , rising slightly from 0.068 to 0.073, while the quantile still falls. Whether the decay seen from  $d = +7$  on reflects some fixed integer's ratio worsening, as against newly entering residues with a large integer representative pulling the population down, is not settled by this measurement. The exhaustive fixed-set test above answers that question only at  $d = +5$ , where no deterioration occurs.

*Remark 4.11* (No monotonicity certificate for the minimum). The companion paper's worst-cylinder cascade-factor theorem certifies a different minimum ratio, for a different integer-covering count, by writing it as a convex combination of the previous level's ratios; this gives monotonicity, and promotes one finite computation to a bound at every higher level [7]. Dividing the  $g_\ell$  recursion (1) by  $\bar{g}_\ell(k)$  gives the analogous expansion

$$\frac{g_\ell(k, a)}{\bar{g}_\ell(k)} = \sum_{j \text{ admissible for } a} \frac{\bar{g}_{\ell-1}(k-j)}{\bar{g}_\ell(k)} \cdot \frac{g_{\ell-1}(k-j, a_{\text{old}}(j, a))}{\bar{g}_{\ell-1}(k-j)},$$

where  $a_{\text{old}}(j, a)$  is the level- $(\ell-1)$  source residue that recursion (1) assigns to  $(j, a)$ , and  $j$  admissible for  $a$  means  $2^{j+1}a \equiv 1 \pmod{3}$ , a condition on  $a \pmod{3}$  alone that selects one parity class of  $j$  out of the  $2 \cdot 3^{\ell-1}$  available. The expansion would be a convex combination, certifying monotonicity the same way, exactly when its coefficient sum (written  $\mathcal{S}_\ell$ , distinct from Wirsching's transition operators  $S_\ell$  of §2)  $\mathcal{S}_\ell(k, a) := \sum_{j \text{ admissible for } a} \bar{g}_{\ell-1}(k-j)/\bar{g}_\ell(k)$  equals 1 for every  $a$ . It does not, and not even on Haar average. Wirsching's bounded-urn count  $q_\ell(k) = 2 \cdot 3^{\ell-1} \bar{g}_\ell(k)$  of §3 satisfies its own recursion on the last urn's occupancy,  $q_\ell(k) = \sum_{n < c_{\ell-1}} q_{\ell-1}(k-n)$ ; dividing by  $2 \cdot 3^{\ell-1}$  gives  $\sum_{j < c_{\ell-1}} \bar{g}_{\ell-1}(k-j) = 3\bar{g}_\ell(k)$  for every  $(\ell, k)$ , the factor of 3 coming from the ratio  $2 \cdot 3^{\ell-1}/(2 \cdot 3^{\ell-2})$  between the unit counts normalizing  $\bar{g}_\ell$  and  $\bar{g}_{\ell-1}$ . Since every  $j$  is admissible for exactly one residue class of  $a \pmod{3}$ , this splits the sum as  $\mathcal{S}_\ell(k, a \equiv 1) + \mathcal{S}_\ell(k, a \equiv 2) = 3$  identically. A convex combination averaging to 1 would need that sum to be 2; the Haar average of  $\mathcal{S}_\ell(k, \cdot)$  over  $a$  is  $3/2$  at every level. At  $\ell = 3$ ,  $k = 3$ , exact evaluation gives  $\mathcal{S}_3(3, a) = 9/7$  for  $a \equiv 1 \pmod{3}$  and  $12/7$  for  $a \equiv 2 \pmod{3}$ ; the same two-values-per-level pattern, forced by the argument above, holds at  $\ell = 4$ :  $\mathcal{S}_4(4, a) = 6/5, 9/5$  and  $\mathcal{S}_4(5, a) = 5/4, 7/4$  for  $a \equiv 1, 2 \pmod{3}$  respectively. The identity's own left side still averages to 1 over  $a$ , by the definition of  $\bar{g}_\ell$  as that average; the convex-combination mechanism that certifies the companion paper's minimum has no analogue here, not even in the mean. The minimum itself confirms the failure directly: at  $k = \ell + 12$ ,  $\min_a g_\ell(k, a)/\bar{g}_\ell(k)$  equals  $9/28, 648/1459, 2430/5057, 13851/30695$  for  $\ell = 4, 5, 6, 7$  (0.3214, 0.4441, 0.4805, 0.4512), rising and then falling.

**Empirical Result 4.12** (Fourier budget for fixed-cost multiplicities). *For  $p_{\ell,k}(a) = g_\ell(k, a)/\sum_b g_\ell(k, b)$ , with  $u_\ell$  the uniform law on unit residues mod  $3^\ell$ , additive Fourier inversion gives*

$$\min_a \frac{p_{\ell,k}(a)}{1/(2 \cdot 3^{\ell-1})} \geq 1 - \frac{2}{3} \sum_{\xi} |\widehat{p}_{\ell,k}(\xi) - \widehat{u}_\ell(\xi)|.$$

*At the central cost  $k = \ell$ , the spectral sum grows from 12.11 at  $\ell = 4$  to 293.29 at  $\ell = 12$  and is dominated by primitive frequencies. At  $k = \ell + \lfloor \sqrt{\ell} \rfloor$ , it grows from 7.61 to 122.63. At  $\ell = 12$ ,*

the largest individual discrepancy is 0.339 at central cost and 0.280 in the upper window, while the largest primitive discrepancies are 0.0246 and 0.00682. Thus the largest coefficients retain a coarse component even as each primitive coefficient becomes smaller. The resulting lower bounds are negative throughout, so termwise absolute summation of the finite Fourier series fails to establish uniform positivity on this range.

## 5 Certified numerical test of Wirsching’s Conjecture 3

Conjecture 3 asserts (★5): there are constants  $c, \delta_5 > 0$  with  $\lim_\ell \varphi(z_\ell)/\varphi_0(z_\ell) = c > 0$  uniformly for sequences  $(z_\ell) \in \tilde{A}_{\delta_5}$  [1, §7]. The source proves (★5)  $\Rightarrow$  (★4) only for  $\delta_5 > \delta$ , strictly larger than the class tested by (★4) itself; we test (★5) below without re-deriving that implication. For  $(x_\ell) \in \tilde{A}_\delta$ , write  $x_\ell^+ := x_\ell + 3^{-\ell-1}$  [1, §7]. The test below uses  $x_\ell = k_\ell/3^\ell$  with  $k_\ell = \ell + \lfloor u\sqrt{\ell} \rfloor$  for  $u \in \{-2, -1, -0.5, 0, 0.5, 1, 2\}$ , the class  $\delta \approx 2$  window announced as “ $u \in [-2, 2]$ ” in Empirical Result 5.1 below.

Using the exact self-similarity  $X \stackrel{d}{=} (2U + X)/3$ , the moments  $M_i = \mathbb{E}[X^i]$  satisfy the exact rational recursion

$$M_i = \frac{1}{3^i - 1} \sum_{k=1}^i \binom{i}{k} \frac{2^k}{k+1} M_{i-k},$$

and, combined with an antiderivative reduction expressing  $\varphi$  at extreme-tail arguments in terms of exact binomial moment sums, we avoid iterating  $W_3$  itself, whose  $L^1$  error certificate is too weak to control pointwise values at the relevant scale. This gives an evaluation of  $\varphi$  at depth  $\ell$  up to  $\ell = 500$  (working throughout in log-space with 100-digit precision) whose truncation error is exactly zero at every point tested below.

**Empirical Result 5.1** (Numerical test of Wirsching’s Conjecture 3). *By  $\varphi'(x) = \frac{9}{2}\varphi(3x)$  on  $[0, 2/3]$  [1, eq. (7.7)],  $L_\ell = \frac{2}{3^{\ell+1}}\varphi'(x_\ell^+)/\varphi(x_\ell^+)$ , the quantity (7.5) of [1] needs; its predicted limit  $2/3$  is (7.13) there. Measured for  $\ell = 200, \dots, 500$  in the CLT window ( $u \in [-2, 2]$ ),  $L_\ell$  converges to  $2/3$  for every  $u$  tested. At the central sequence  $u = 0$ , the deficit  $\ell \cdot (2/3 - L_\ell)$  stabilizes near 0.802 (0.8026 at  $\ell = 200$ , 0.8021 at  $\ell = 500$ ). This convergence of the deficit is not uniform in  $u$ : at  $\ell = 500$  it ranges from  $-32.1$  at  $u = -2$  to  $28.3$  at  $u = 2$ , since  $\ell \cdot (2/3 - L_\ell)$  isolates a  $u$ -dependent correction of order  $\sqrt{\ell}$  away from the central sequence. The quantity  $\ln(\varphi/\varphi_0)$  itself, by contrast, is close to uniform across the seven values of  $u \in [-2, 2]$  tested at fixed  $\ell$ : at  $\ell = 500$  the spread is  $6 \times 10^{-5}$ . Along  $u = 0$ ,  $\ln(\varphi/\varphi_0)$  is increasing and concave in  $\ell$ , consistent with a finite limit; a least-squares fit to  $L_\infty + a/\sqrt{\ell}$  (distinct from  $L_\ell$  above) over  $\ell = 200, \dots, 500$  gives  $L_\infty = -0.619$ ,  $a = -0.79$ , hence  $c = e^{L_\infty} \approx 0.539$ . Fitting only the upper half of the range ( $\ell \geq 350$ ) moves  $L_\infty$  by under  $10^{-4}$ ; adding a further  $1/\ell$  term moves it by  $4 \times 10^{-4}$ , the wider of the two checks. The zero-truncation-error certificate above bounds the evaluation of  $\varphi$  itself, not this extrapolation. Conjecture 2 remains open; Theorem 3.1 settles Conjecture 1 independently of this computation.*

*Remark 5.2* (A same-phase check against a periodic factor). Conjecture 3 asks for a constant  $c$ ; Berg and Krüppel’s own expectation is weaker, a ratio  $\varphi/\varphi_0$  bounded and bounded away from zero [2, §9], which a nonconstant  $\varphi = Q \cdot \varphi_0$  with  $Q(\log_3 x)$  1-periodic would also satisfy.  $L_\ell$  cannot distinguish the two, for a reason needing no special construction:  $\log_3(3x) - \log_3(x) = 1$  exactly for every  $x > 0$ , so  $Q(3x) = Q(x)$  identically, and  $L_\ell$ , a ratio of  $\varphi$  at  $3x_\ell^+$  and at  $x_\ell^+$ , is blind to any such  $Q$  everywhere it is evaluated.

A different, sharper test asks whether  $\ln(\varphi/\varphi_0)$  is a smooth function of  $\ell$ , as the deficit’s own  $1/\sqrt{\ell}$  fit supposes, or carries extra structure tied to the phase of  $x$  in  $\log_3$ , as a nonconstant periodic

$Q$  would require. At the bare points  $x_\ell = \ell/3^\ell$ , not  $x_\ell^+$ , the pair  $(\ell, 3\ell)$  shares phase exactly:  $x_{3\ell}/x_\ell = 3^{1-2\ell}$  is an exact rational identity (verified for every tested pair with no rounding), so  $Q(x_{3\ell}) = Q(x_\ell)$  for a 1-periodic  $Q$ , and if  $\varphi_0$  were exact to all orders, the drift  $\ln(\varphi/\varphi_0)(x_{3\ell}) - \ln(\varphi/\varphi_0)(x_\ell)$  would vanish identically at every such pair. It does not, but the drift moves toward the value the smooth fit itself predicts as  $\ell$  grows: observed drift  $-0.013, +0.039, +0.032, +0.027$  against fit-predicted  $+0.106, +0.047, +0.034, +0.027$  at  $(\ell, 3\ell) = (10, 30), (50, 150), (100, 300), (150, 450)$ , a relative mismatch of 112%, 18%, 4%, and 1%. A fixed periodic contamination would leave a mismatch of fixed size at every depth; here the mismatch shrinks with  $\ell$ , matching a finite-size correction to  $\varphi_0$ . This disfavors a dominant periodic factor without ruling out a small one, nor proving  $c$  constant.

## 6 Discussion

The mass decomposition (2) and the equivalence and nonequivalence theorems above are exact; they bound how far local-limit methods reach toward Conjecture 2. Theorem 4.4 and Theorem 4.5 give convergence at fixed and sublinear precision, checked at finite level by Empirical Result 4.8; Empirical Result 4.12 finds that termwise absolute summation of the finite Fourier series gives only a negative, vacuous lower bound at every level tested. Theorem 4.6 shows the latent cost vectors provably fail to converge at linear precision. Theorem 4.7 is a separate, one-sided bound at full precision,  $D_\infty \leq \frac{1}{2} \log \ell + O(1)$ ; it does not explain the linear-precision failure. It only upper-bounds the likelihood-ratio gap, in the direction opposite to the lower bound condition (★3) actually needs. Condition (★4), the hypothesis Conjecture 2 poses toward (★3), carries no information about the coordinate this machinery controls (§2). Closing the gap needs an argument on  $W_3$  itself, or a bridge between (★4) and (★3) uniform in growing resolution. Neither bridge the source supplies reaches that far.

The open hypothesis in Theorem 4.1 differs from the equivalence-of-ensembles results above: it is a uniform lower bound on the generators themselves. Empirical Result 4.10 bears on it two ways. At the support boundary  $d = +5$ , an exhaustive test over a fixed, non-growing set of integers shows no deterioration through  $\ell = 17$ . At  $d = +6$  the population minimum, over the full growing set of units, does not decay either; the quantile already does. From  $d = +7$  on, the population’s low quantiles decay at a rate close to the minimum’s own, even though the quantile itself sits well above the extreme order statistic, and whether that decay reflects some fixed integer’s ratio worsening is untested at those offsets. Remark 4.11 closes off the one available shortcut regardless: the level-to-level monotonicity certificate that works for a different integer-covering count in the companion paper does not transfer, with an explicit non-monotone counterexample.

The numerical test of Conjecture 3 (§5) is exact where it applies, but  $L_\ell$  alone cannot separate a limit constant from a periodic factor; Remark 5.2 adds a same-phase drift check that disfavors a dominant periodic factor, without separating the two fully.

## Data Availability Statement

Code and data reproducing every claim in this paper, organized by section with a README per subfolder, are at <https://github.com/faculdade/collatz-wirsching-2003>.

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